DHAIRYA BHATIA

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EDUCATION

Khoury College of Computer Science, Northeastern University — Boston, MA
Master of Science in Computer Science — Sep 2025 – May 2027 (Expected)

The University of British Columbia — Kelowna, BC
Bachelor of Science in Computer Science — Sep 2020 – May 2024

RESEARCH EXPERIENCE

Augmented Cognition Lab (ACLab), Northeastern University — May 2026 – Present
Research Assistant · Advisor: Prof. Sarah Ostadabbas

- Co-led MotionBlind, a contrastive benchmark of 60 video pairs (82 self-recorded clips) that differ only in speed, magnitude, or direction.
- Found open Video-LLMs score 52 to 59% on single questions but at most 11.7% on full pairs, against 91.3% for humans.
- Built SLURM evaluation pipelines on HPC to benchmark 8 Video-LLMs (6 open-weight, 2 closed-source) across 3 benchmarks.
- Showed that shuffling, reversing, or removing frames drops open models to near 0%, proving the task requires ordered video.
- Compared 4 frame-selection strategies across 1 to 24 frame budgets and found none beat uniform sampling.
- Collecting an egocentric and exocentric video dataset for multi-view video understanding.

CACTILab, Northeastern University — Mar 2026 – Present
Research Volunteer

- Evaluated multiple C-to-Rust translation tools.
- Built a multi-agent LLM pipeline that generates embedded firmware and verifies it with the compiler in the loop.
- Delivered a working firmware MVP generated by the pipeline and tested on a TI MSPM0L2228 board.
- Added model tiering and per-turn cost tracking to control LLM usage across agents.

PUBLICATIONS

Dhairya Bhatia, Bishoy Galoaa, et al. *MotionBlind: Probing the Illusion of Motion Understanding in Video-LLMs*. arXiv:2609.09528, 2026. **Co-first author.**

INDUSTRY EXPERIENCE

Oriv · Software Development Intern / AI Developer — Sep 2024 – Jul 2025

- Prototyped knowledge-graph methods for fitting graph structures onto IoT sensor signals for industrial automation POCs.
- Introduced sensor-virtualization methods enabling data simulation and reducing dependency on physical hardware.

PROJECTS

ECG Diagnostic Classification · PTB-XL

- Trained a 1D ResNet in PyTorch on PTB-XL, reaching 0.922 macro AUROC across 5 diagnoses over 3 seeds.
- Showed shuffling whole beats cuts AUROC by only 0.004, while breaking beats apart cuts it by 0.076.
- Tested on the external Georgia ECG dataset without retraining, where macro AUROC fell to 0.835.

NYC Taxi Duration and Tip Prediction

- Compared 5 models from a group-median baseline to LightGBM and a PyTorch MLP on NYC TLC trip records.
- Cut trip-duration test MAE from 4.24 to 2.77 minutes (35%) with LightGBM, raising R^2 from 0.65 to 0.84.
- Raised tip-prediction AUROC from 0.682 to 0.747 over a zone-rate baseline, with calibration error of 0.002.
- Showed every model degrades on the following month, with duration MAE rising by up to 0.23 minutes.

Project Escalion · NU Launch Labs (Co-Founder & Team Lead)

- Co-founded a Rust game development studio that grew into a Khoury-recognized graduate club.
- Building a custom Rust engine with a wgpu renderer and a sparse-set entity component system.

TEACHING

Teaching Assistant, CS3100, Northeastern University — Jan 2026 – Jun 2026

- Supported students in labs and office hours.
- Led analysis of student feedback on AI-assisted programming workflows, informing curriculum design.

TECHNICAL SKILLS

ML / AI: PyTorch, Multimodal Learning, VLMs, Model Evaluation & Benchmarking, Computer Vision
Languages: Python, Rust, C, Java, SQL, Bash
Systems & Tools: Linux, SLURM, HPC, Docker, Git, Embedded Systems